Nursing Practice Guideline

Treatment of Burns

Reference #: NPG 27

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Nursing

Introduction

This guideline reflects the best available evidence in the assessment and treatment of burns at SCGH and Osborne Park Hospital group.

Burns Terminology:

- A burn injury is damage to the skin and deeper structures as a result of flame, hot liquids, contact with hot or cold objects, electricity and chemicals.
- Total body surface area (TBSA) relates to the percentage of the body that is injured by the burn. Refer to Diagram 1: Wallace ‘Rule of nine’. Adult burn injuries are separated into minor burn <15% TBSA or major burn > 15% TBSA. Major burns require transfer to the specialist burns service at Fiona Stanley Hospital.
- Depth of injury relates to superficial, partial or full thickness injury. Refer to Appendix 1.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☒	Staff Safety	☒
Other:	☐		

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Principles

Observe the five moments for hand hygiene. Follow aseptic technique and use the risk assessment outlined in the SCGOPHCG Infection Prevention and Management Aseptic Technique Policy. This must include use of appropriate Personal Protective Equipment (PPE) to prevent contamination and mucosal or conjunctival splash injuries[5].

1.1 Medical Requirements[2]

Patients should be transferred to the burns unit at Fiona Stanley Hospital (FSH) with:

- Burns of greater than 15% Body Surface Area (BSA) (see Diagram 1) and/or burns >5% full thickness (see Appendix 1).
- Burns in particularly important areas such as face, hands, feet, and perineum or over a major joint.
- Circumferential, infected or electrical burn.
- Burns with concurrent injuries or co-morbidities.
- Chemical burns.

If patient transfer to Fiona Stanley Hospital is anticipated, and transport time is **more than 2 hours** cover the wound surface with antimicrobial dressings (note: Acticoat™ is the preferred dressing, apply Acticoat™ dressings moistened with sterile water, followed with sterile water compresses and dry gauze)[3].

SCGH Plastic Surgery Team should be notified if a patient with a burn presents to ED or is an inpatient.

SCGH Plastics Surgery Team will give specific orders for burn/wound care depending on type of burn and burn history.

1.2 Safety Information

Patients with burns are particularly at risk of infection therefore:

- Admit the patient into a single room, where appropriate (e.g., immunocompromised).
- Practice standard precautions.
- Assess the need for tetanus prophylaxis[4].

If the injury is a burn to the hand, ensure all rings and jewellery are removed; a burn to the hand may affect circulation. Rings / Jewellery removal kits can be obtained from:
OPH: Operating Theatre
SCGH: Emergency Department or Intensive Care Unit.

1.3 Nursing Management [4]

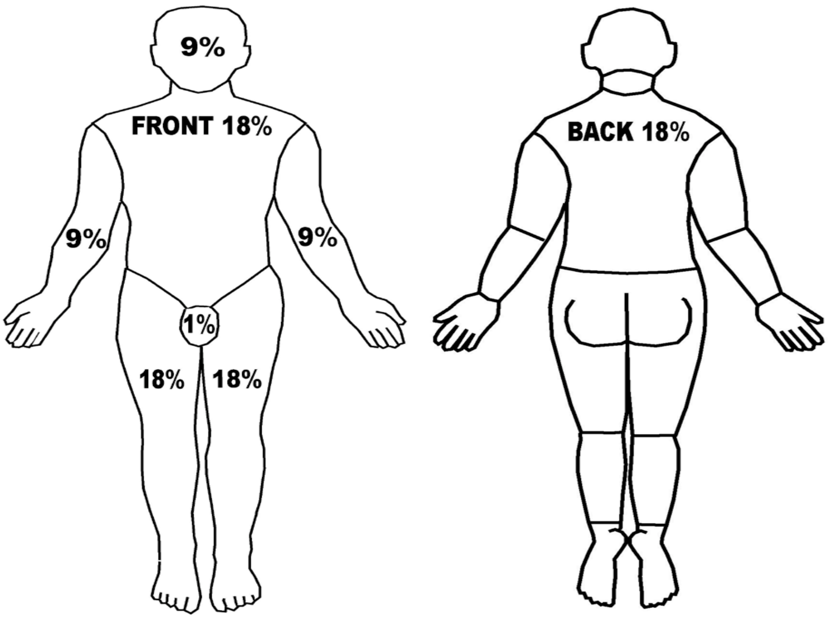
Initial patient assessment and burn history will influence nursing management within the first 24 hours.

Consider:

- **First aid** given at the time of injury.
- **Airway** – check patency, particularly a burn to face or neck which is a result of explosion.
- **Breathing** – any respiratory distress, hoarseness, stridor, voice changes.
- **Circulation** – hydration and fluid requirements, capillary refill, circumferential burn, need for escharotomy.

- **Depth** of injury (superficial, partial or full thickness injury). Refer to Appendix 1: Depth Characteristics of Burn Wounds.
- The anatomical **location** of the burn,
- Presence of **oedema** or **blistering,**
- **Pain.**
- Burn **size** (BSA %): calculate Percentage of Total Body Surface Area (TBSA) for partial and full thickness burns[4]- refer to Diagram 1: Wallace 'Rule of Nines'.

Diagram 1: Wallace 'Rule of nines' diagram[6]



Body / Area of Burn	**Percentage Equivalent to TBSA**
A person's palm	1%
Whole hand	9%
Whole arm	9%
Whole leg	18%
Front trunk	18%
Back trunk / buttocks	18%
Perineal / genitals	1%

Burn History:

- Patient confined in an open or enclosed space (risk of inhalation injury),
- Explosion/blast, other injuries associated with burn,
- Chemical type,
- Time before first aid was commenced and what type of first aid was given at the scene,
- Identify cause of burn: flame, scold, chemical or electrical,
- Time of injury.

Medical History:

- Allergies,
- Pre-existing diseases or co – morbidities,
- Past medical history,
- Check the patient for drug and current medications.

Initial treatment:

- Cool the burn. If the burn has just occurred, initial treatment should include cooling the burn under cool running water, or by applying continuous cool compresses for at least 20 minutes within the first 3 hours post burn (No Ice).
- Pain management. Superficial and partial thickness burns can be painful so ensure analgesia is administered as required.
- Contact Plastics Registrar On call.

Microbiology:

- Take swabs from the burn/s if there are clinical signs of infection present.

Cleansing and Dressing Guidelines of minor burn [5,6]

- Using a non-touch technique, cleanse the burn area with Chlorhexidine 4% pre op wash and tap water.
- Initial assessment of the burn according to depth of injury-superficial, partial or full thickness. Refer to Appendix 1: Depth Characteristics of Burn Wounds Guideline.
- Dressing options will be dependent on assessment of the individual burn wound including size, exudate level, type of tissue, depth of injury, presence of infection, peri-wound skin condition and location. Refer to Appendix 2: Dressing Guidelines Table.
- In the instance of minor burns, it is recommended that the burn be reassessed in the first 24hrs to determine the depth of injury (avoid using an occlusive adhesive dressing). Burns are dynamic and the injury can continue to extend.

1.4 Procedural Information

Refer to Plastic Surgeon / Surgeon specific orders.

See attached Appendix 2 for specific dressing recommendations.

Ongoing Management / Follow-up:

If the patient is discharged with an existing burn wound and is under the care of the Plastics Team SCGH follow up will be in the plastics dressing clinic. All other minor burns are to be referred to FSH Burns outpatient clinic.

Related Documents

Legislation:

- Health Practitioner Regulation National Law (WA) Act 2010
- Nursing and Midwifery Board of Australia (NMBA) Code of Professional Conduct for Nurses March 2018
- Work Health and Safety Act 2020.

Policy:

- SCGOPHCG Infection Prevention and Management Aseptic Technique Policy
- SCGOPHCG Clinical Handover - Communicating for Safety
- SCGOPHCG Procedure Matching Policy

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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.

Appendix 1: Depth Characteristics of Burn Wounds

	1.5 Superficial (First Degree)	**1.6 Partial Thickness** (Superficial Second Degree)	**Deep Dermal Partial Thickness** (Deep Second Degree)	**1.7 Full Thickness** (Third Degree)
Predisposing cause	Sunburn; ultraviolet exposure; minor flash	Brief exposure to flash flame; scalds; brief exposure to dilute chemicals	Hot liquids or solids; flash flame, direct flame; intense radiant energy; chemicals	Prolonged contact with flames, hot liquids, steam; chemicals; high voltage electric current
Depth of injury	Minor epithelial damage Skin not broken.	Epidermis, minimal damage to dermis, Skin is broken, with a wet, red or pink wound bed.	Entire epidermis and more dermal involvement than superficial partial thickness; intact hair follicles and sweat glands.	Epidermis, dermis, epidermal appendages; portion of subcutaneous fat, possible involvement of connective tissue, muscle, or bone
Physical characteristics	Red or light red, dry or small blisters; slight erythema; exquisitely painful to touch	Moist, bright pink or red colour; blister formation; intact blanching; tactile and pain sensation, capillary refill time is brisk.	Pale waxy appearance; absent blanching; mostly dry; decreased pinprick sensation but pressure sensation intact, capillary refill time is slow.	Dry, leather, insensate, avascular, pale yellow to brown in colour, possibly charred; thrombosed vessels, non-blanching, capillary refill will be absent.
Healing time	Approx. 3 - 5 days	Approx. 7 - 14 days	Prolonged healing period > 14-21 days; contracture formation; possible conversion to full thickness injury; hypertrophic scarring. May require grafting	Incapable of self-regeneration; requires grafting unless very small

Appendix 2: Dressing Guidelines

Burn wound characteristics	Generic group/*Trade name*	Advantages	Disadvantages	Frequency of dressing change
▪ Superficial injury / Erythema only e.g. sunburn, flash burn	Moisturiser *e.g. Sorbolene* Emollient ointment Retention dressing (as below) [10]	• Soothing	• Requires frequent application	• **At least 3-4 times/day**
• Superficial to partial thickness injury • Small amounts of exudate	Hydrocolloid dressing[10] *e.g. Duoderm CGF,* *Adhesive Allevyn/ Mepilex foam border* [11,12] *Mepilex/foam*[11,12] *Acticoat covered with hydrocolloid*	• Occlusive • Promotes autolytic debridement. • Promotes granulation. • Self- adhesive • Reduces pain. • Antimicrobial action: prophylaxis for infection AND will reduce existing bacterial burden	• Not transparent • May produce offensive odour. • May leak and macerate skin	• Prior to exudate leaking • 2-5 days
• Partial thickness injury • Minor bleeding wounds	Calcium alginate[13] *e.g. Curasorb, biatain Alginate*	• Haemostatic • Absorbs exudate. • Reduce pain	• Ineffective on dry wounds/eschar • Requires a secondary dressing	• Prior to exudate leaking • Donor site dressings can be left intact 10-14 days
• Moderate to large exudate	Calcium alginate with silver *e.g. Melgasorb* [Ag]	• Mildly haemostatic • Absorbs exudate	• As above	• Prior to exudate leaking • Leave intact 2-7 days
• Partial to full thickness injury – small areas (i.e. < 5cm^2) • Presence of slough • Small to moderate exudate • Non-infected wound	Hydrocolloid e.g. Duoderm CGF, or Allevyn Adhesive with Fixomul applied as a border Mepilex foam border **OR** Acticoat Flex covered with hydrocolloid	• Occlusive • Promotes autolytic debridement. • Promotes granulation. • Self- adhesive • Reduces pain	• Not transparent • May produce offensive odour. • May leak and macerate skin	• Prior to exudate leaking • 2-5 days
• Deep Partial to full thickness injury • Infected, dirty or critically colonised burn	Acticoat / sterile water [3, 14] compress (2 layers of wet gauze and two layers of dry gauze to activate Acticoat) If clinic is closed, then Acticoat flex (Day) under Hydrocolloid e.g., Duoderm CGF.	• Softens eschar. • Reduces incidence of burn infection • Soothing	• Mildly cytotoxic • May cause stinging on superficial burns. • Requires secondary dressing	• Daily change essential, frequency of dressing changed to be prescribed by Plastic Surgical team. • Pt must be sent to the burns clinic.